

Cesare ROSSI, Flavio RUSSO, Sergio SAVINO

Windmills: Ancestors of the wind power generation

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Abstract A brief description of the windmills from the second millennium BC to the Renaissance is presented. This survey is a part of several studies conducted by the authors on technology in the ancient world. The windmills are the first motor, other than human muscles, and are the ancestors of the modern wind turbines. Some authors' virtual reconstructions of old windmills are also presented. The paper shows that the operating principle of many modern machines had already been conceived in the ancient times by using a technology that was more advanced than expected, but with two main differences, as follows: Similar tasks were accomplished by using much less energy; and the environmental impact was nil or very low. Modern designers should sometimes consider simplicity rather than the use of a large amount of energy.

Keywords windmills, ancient machines, history of windmill

1 Introduction

This research is a part of the several studies on the ancient technology in general [1–3] and on the machines of the ancient world [4–7]; on this topic, namely history of mechanism and machine science, the authors have carried out studies for many years. Moreover, the ancient windmills were studied by few authors [8–20].

The first non-manual source of energy used by human was wind energy. Examples of wind motors are, in fact, very old. Wind motors are all devices that supply energy by using the kinetic energy released by the movement of an aeriform mass. Man has learned to harness the strength of

the atmospheric currents since the ancient times, as evidenced by the development of the sails for ship propulsion. The application of the wind power had many children, including the windmill. Windmill was a normal and a widespread reality at the time of the great Hammurabi who reigned between 1792 BC and 1750 BC.

Although the advent of the windmill was believed to be in the twelfth century BC in the green Dutch meadows (hence, it is much modern than the water mill), its invention was remote. From a chronological point of view, the impeller rotated by the wind made its debut on the threshold of history in Mesopotamia.

The Sanskrit has the adjective *tur-as* and the verb *tur-ami*, which respectively signify fast and to speed up. The dynamic meaning of the word *tur* is implicit. It was first obtained from the Latin and then from the Italian. It acquired the much forceful meaning of fast and whirling motion and of rotary motion as for cyclones and whirlpools such as turbine, tornado, torment, and perturbation. By figurative analogy, the perturbation and disturbance are all synonymous with a sudden and radical inversion of the state of being.

Turbine, the physical reality of the etymological root word, was first defined simply as paddle wheel or, in respect of its primary use, windmill. Regarding this last accepted meaning, specifically indicating the motor of a machine used for grinding, they soon had to specify whether wind or water was the natural currents used.

In the paper, some of the authors' virtual reconstructions of old windmills are also presented. All these are represented in isometric projection. A scale to roughly evaluate the dimensions is also reported.

2 Vertical axis windmills

At this point, the design of the wind device that turned the wheels of Hammurabi is not yet determined. In the same region, around the seventh century BC, a singular wind impeller appeared showing a rational conception that represents the most archaic prime mover. Several clues lead to a conclusion that it was similar to much recent

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Cesare ROSSI (✉), Sergio SAVINO
Department of Industrial Engineering University of Naples "Federico II"
via Claudio, Naples 80125, Italy
E-mail: cesare.rossi@unina.it

Flavio RUSSO
Consultant of Ufficio Storico Stato Maggiore Esercito, Torre Del Greco
80059, Italy

Chinese windmills used for irrigation that have vertical axis blades [8–13]. This kind of windmill, which is extremely simple, does not require strong and constant winds because it can work with light and variable breezes. It was essentially made by a type of carousel poles connected to each other and to a central vertical shaft.

In detail, it consists of two equal horizontal frames, upper and lower, having eight or six sides, each one about 3 to 4 m long; from each edge a radial strip departs then fits in the shaft, and vertical struts about 3 m long, which connect the frames. To each strut, a mat is fixed that can orientate because of the oblique arrangement of the poles in the lower frame, thereby offering the maximum aerodynamic resistance in one direction and the minimum in the opposite. In Fig. 1, the authors' virtual reconstruction of the windmill described above is shown, based on several sources [8,12,13].

2.1 Afghan windmill

Over time, the diameter of the impellers was gradually decreased and the sails were made of increasing height, which fitted directly on the shaft. This became the rotor of the forerunner turbine, which exposed only a few blades to the wind through a long and narrow slit made in the tower that contained it. This policy is still adopted to convey the fluid on the blades of the turbines.

The archetype of the wide family of vertical axis mills is the Afghan mill or the Persian mill. It consists of a shaft provided with various mats radially disposed and acting as blades, to the lower end of which a horizontal grinder stone

was wedged and fixed over another identical one. The impeller was inserted into a building, on top of which two closely spaced beams originate a rudimentary strain for the upper end of the shaft, ensuring the vertical alignment and the rotation.

An ancient source described them as follows: “They have eight wings behind two pillars between which the wind has to push a wedge. The wings are mounted vertically on a vertical pole the lower end of which moves a grindstone that rotates above another one below” [20]. The two pillars formed were slightly less wide than the mat but of equal height, through which the wind penetrates in a constant direction.

In Fig. 2, the authors' virtual reconstruction of the above-described Afghan windmill is shown, and in Fig. 3, a technical drawing of the same is also presented.

2.2 Vertical axis windmill in the Renaissance

The actual period during which the vertical axis windmill reached Europe in the ancient times was unknown, but its existence in relatively modern times is certain. This can be seen in a drawing in 1595 by Fausto Veranzio (1551–1617), which is shown in Fig. 4 [21].

The real novelty, and perhaps the greatest contribution by Veranzio, involves the use of fixed blades with the same height as the rotating blades. The rotating blades were located at a precise angle so that the air flow was conveyed to the rotation group with constant angle of incidence, regardless of the wind direction. The idea, which was not used in the immediate following centuries, reappeared in

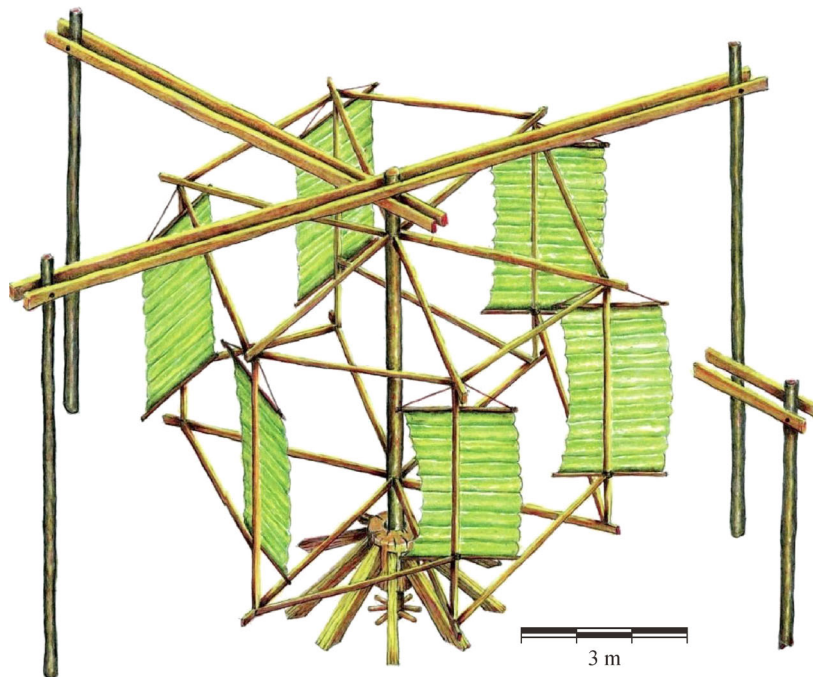


Fig. 1 Virtual reconstruction of the vertical axis windmill

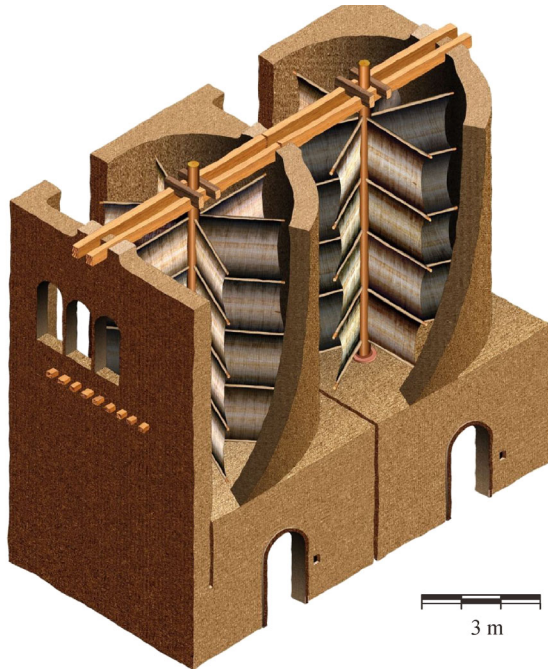


Fig. 2 Virtual reconstruction of the Afghan windmill

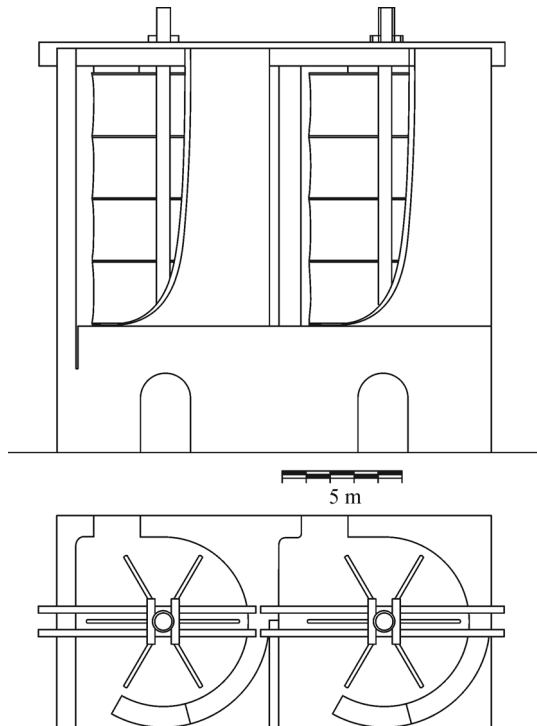


Fig. 3 Drawing of the Afghan windmill

the Francis turbine, where a crown of fixed blades directed several highly strong jets of water on the rotor blades improving the performance. Figure 5 [21] is a drawing by Veranzio showing the fixed blades.

3 Horizontal axis windmills

This windmill has become widespread in Europe since the ancient times. Its constructive solution was derived from the fact that in Europe, and especially in the Mediterranean, the wind direction is not constant. Therefore, the rotor has to be oriented so that it is put in the most advantageous way with respect to the wind direction.

3.1 The Cretan windmill

The Cretan windmills were used along the Mediterranean coast until the last century, and can be considered as the ancestors of all the windmills with horizontal axis. However, the singular testimony of Heron [22], a person of such unquestionable competence, became a determining factor. In “speaking of a pneumatic part activated by a wheel with a paddle,” he described the wheel as somewhat similar to an “amenurion” (ἀνεμουριον), which he evidently considered an object well known to the reader. The word consists of a first term that means “wind,” but the context makes it clear that he is speaking of an object capable of creating a rotary motion using the wind. The word ἀνεμουριον, is also a toponym of two promontories in Cilicia. One may conjecture that in this case they were windmills (unless the word is being used to indicate a windy hill, that only by coincidence coincides with Hero’s term). Because all promontories in Greece, and in other countries, are always ventilated, the reference to wind is logical only if it relates to a distinctive element, such as a windmill. A Cretan windmill has an oblique axis, ropes that maneuver, and triangular sails. This is evidently not a confirmation, but rather a significant clue that the mastery of the mechanical skills necessary for such a rotor is applicable to nautical skills from the first marine supremacy in the Mediterranean, the legendary Minoan civilization. The use of different sails around an axis was brilliant, and it was very plausible in a culture that was characterized by the figure of Daedalus.

Structurally, the Cretan wind rotor was made by four to twelve canvas triangular wings, similar to what was used as the sails of a ship. These sails, fixed to a rough wooden structure, were suspended at an angle of about 10° with respect to the level of the rotor, so that they were oblique to the wind. By adjusting the exposed surface, as done on the ships, the torque on the shaft and the rotational speed increased or decreased.

In Fig. 6, the authors’ virtual reconstruction of the Cretan windmill rotor is shown.

The power supplied by a Cretan windmill varies according to the number and the size of the sails [23]. Having a number of small windmills was preferred than having a few large ones because they were easier to maneuver and operate. For the same reason, the rotors with only four or at most six wings were realized, relegating the

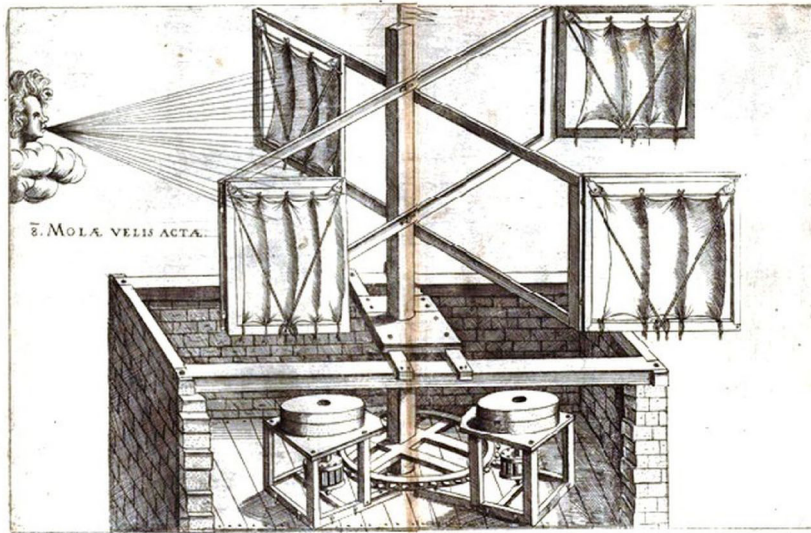


Fig. 4 Windmill designed by Veranzio [21]

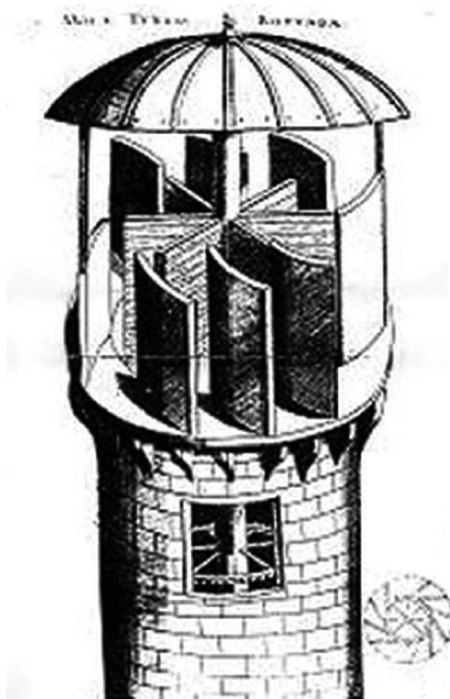


Fig. 5 Windmill by Veranzio with fixed stator blades [21]

rotors with a larger number of sails to much difficult tasks or less windy places.

Cretan windmills were still used in the Mediterranean until recently. In Fig. 7, a Cretan windmill still existing in modern times is reported.

3.2 More recent horizontal axis windmills

The great stimulus that navigation had since the Middle Ages encouraged the expansion of longer trade routes. The

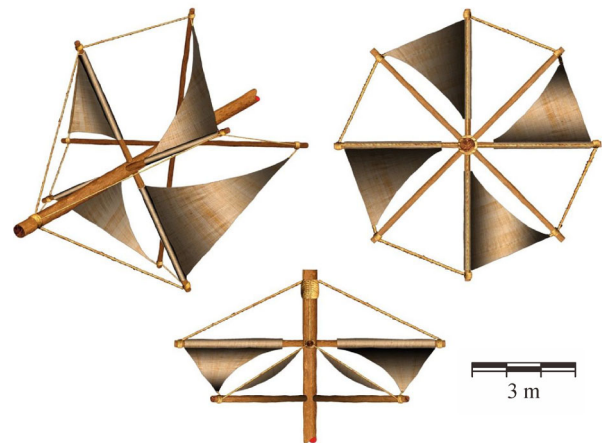


Fig. 6 Virtual reconstruction of the Cretan windmill rotor

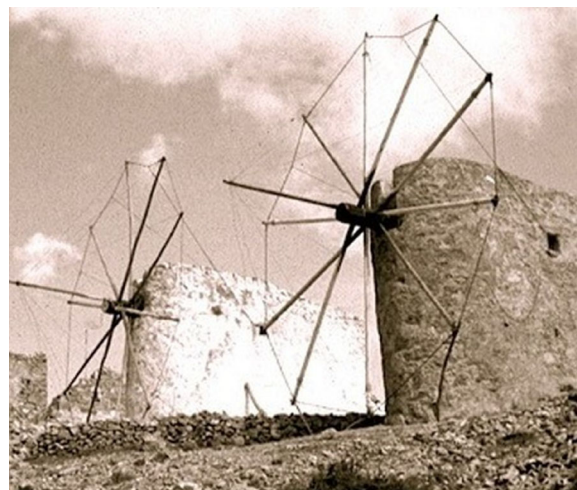


Fig. 7 Windmill in Crete, courtesy of Dr. Antonio Ulzega

wind power can be exploited by oblique directions if the sails were properly made and oriented. Moreover, the Chinese discarded the sails, which were almost rigid and extremely much efficient than European ones (especially with low angles between the wind direction and the sail), and suggested the use of rigid blades for the wings, as in the Dutch windmills.

In Figs. 8 and 9, the authors' virtual reconstructions of the horizontal axis windmill of medieval times are shown. Fig. 8(a) shows the virtual reconstruction of a central pillar mill. The slight inclination of the impeller shaft, the pulley, the winch, and the rear access ladder that ran together at the mill was evident. Figure 8(b) shows the front view of the virtual reconstruction of an archaic central pillar windmill as seen from the rear side of the entrance. The roof is gabled, similar to that of the contemporary housing. In Fig. 9(a), the cross section of the virtual reconstruction of the central pillar mill is reported. It illustrates the mechanism that transforms the rotation of the blades in the grindstone and the winch drive. Figure 9(b) shows the virtual reconstruction of the internal mechanism of the windmills, both the pillar and the tower. In particular, the brake on the wheel rim of the large toothed wheel and the impeller shaft can be observed.

4 Rough evaluation of the power from an ancient wind motor

The power (P) that will be obtained from the wind motor of a windmill can be roughly computed as follows:

$$P = \frac{1}{2} \rho A_r V^3 \eta$$

where V is the air speed, ρ is the air density ($\sim 1.25 \text{ kg/m}^3$), A_r is the rotor surface perpendicular to the current, and η is the coefficient of efficiency.

For the ancient windmills, the following indicative values of η can be assumed: Vertical axis windmill: $\eta = 0.1$; Cretan windmill: $\eta = 0.25$; traditional mill: $\eta = 0.25$ to 0.3 . Therefore, considering that a Cretan windmill has a 6 m diameter rotor, the powers will be roughly developed depending on the wind speed, as reported in Table 1.

The power in Table 1 may seem very low, but is actually high. In fact, that power must be compared with the power available from manual energy. The average power provided by a man working continuously on a crank for long periods is about 76 W and around 230 W for short periods; a horse with a mass of 400 kg will exert a traction force of 400 N. When the horse exerts such a traction while proceeding at a speed of 0.9 m/s, the (continuous) power provided will be equal to $400 \times 0.9 = 360 \text{ W}$ [24]; similarly, about 225 W can be computed for a donkey with 250 kg mass, and about 500 W for an ox proceeding at 0.6 to 0.85 m/s [24].

Therefore, a Cretan windmill with a 6 m diameter rotor can satisfactorily substitute a horse working in a mill.

5 Bearings

Not much information is available on bearings, especially as far as the most ancient ones is concerned. Most

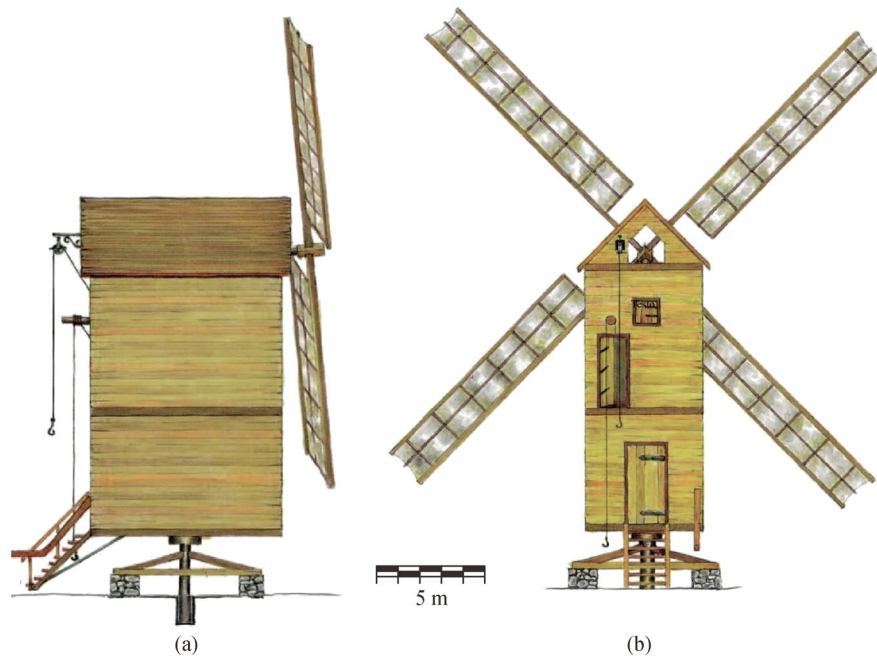


Fig. 8 Virtual reconstruction of a horizontal axis windmill. (a) Side view; (b) front view

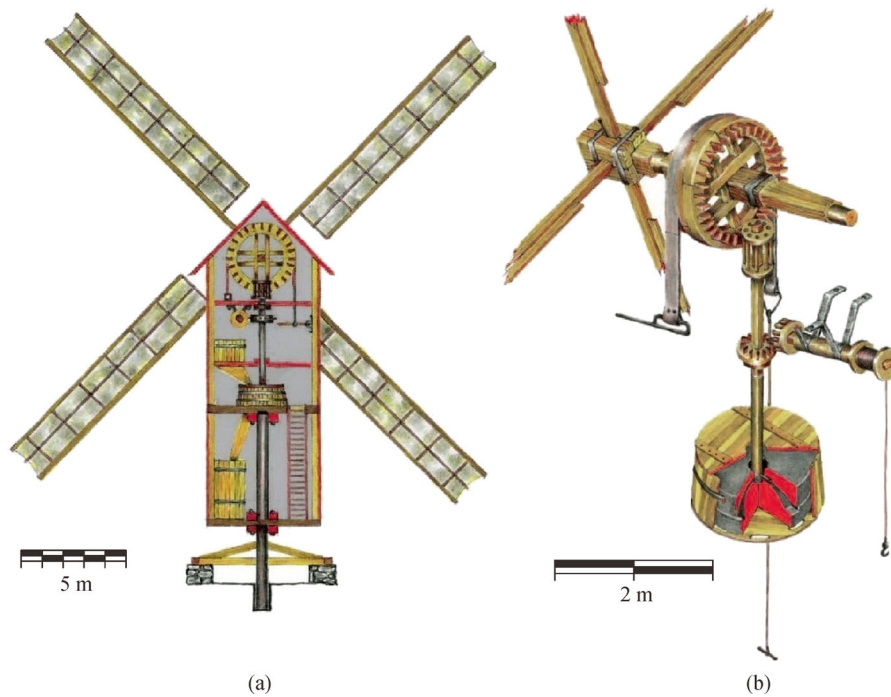


Fig. 9 Virtual reconstruction of the mechanism. (a) Global view; (b) detail

Table 1 Power from a 6 m diameter Cretan windmill rotor

Wind type	Wind speed/(m·s ⁻¹)	Power/W
Light air	1	4.5
Light breeze	3	120.0
Gentle breeze	5	552.0
Moderate breeze	7	1515.0

probably, the wooden shafts of the horizontal axis rotors were housed with two pieces of full bearings also made by hard wood and lubricated by the animal fat. The lower hand shaft of the vertical axis rotors, was probably conical (truncated) and fitted in a stone housing, also lubricated by the animal fat.

Roman charts have iron shafts and an iron bushing fitted in the wheel hubs, and much advanced solutions were adopted in antiquity. In Fig. 10, some examples of both radial bearings and thrust bearings in the Classic Era are shown.

In Fig. 10(a), the authors' virtual reconstruction of a roller bearing of the hub of a Celtic cart is shown. In Fig. 10(b), a drawing of the thrust ball bearing of the spherical platform installed in one of the Nemi ship is shown. In Fig. 10(c), bearing hinges and thrust plates in bronze, which are found at Herculaneum, are shown.

To the authors' best knowledge, no evidence of these much advanced constructive solutions has been found in the ancient windmills in Europe or in the Middle East. Hence, it can be considered as mere possibility.

6 Conclusions

A brief description of the windmills, from the second millennium BC to the Renaissance, was presented with some virtual reconstructions of the old windmills.

Sail windmills were not only the first non-manual source of energy that was invented and a very interesting product of the human ingenuity but can also represent possible sources of energy in the developing countries because of their simplicity.

The operating principle of many modern machines has already been conceived in the ancient times by using a technology that was much advanced than expected. However, two main differences were evident: Similar tasks were accomplished using much less energy; the environmental impact of these plants was nil or very low.

The authors hope that this research could contribute to increase the interest in the history of mechanism and machine science, and it suggests that modern designers consider simplicity rather than the use of large amount of energy.

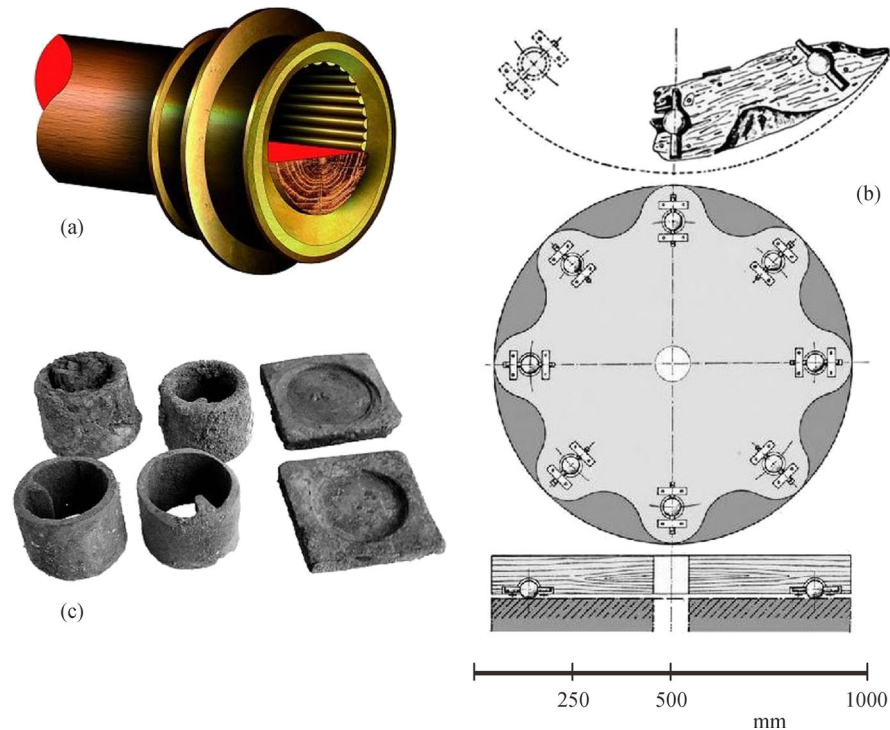


Fig. 10 Examples of ancient bearings, reproduced from Ref. [1]. (a) Virtual reconstruction of a roller bearing; (b) drawing of the thrust ball bearing of the spherical platform; (c) bearing hinges and thrust plates in bronze

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